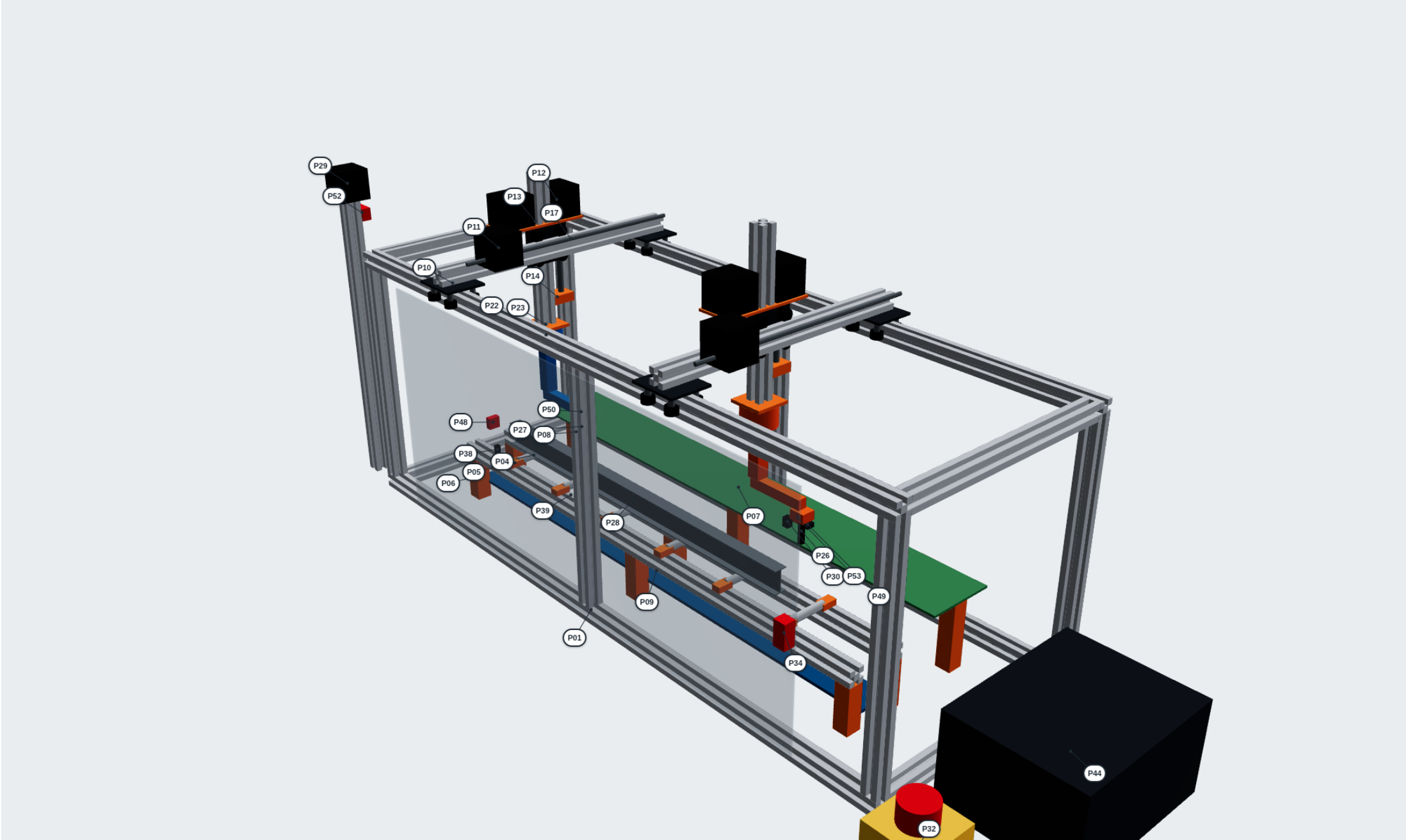


General assembly

Each balloon is a part number (P01 ...) - the same number as the shopping list and the CAD. The table lists every part in the model and how many there are.



Part	Name	No.
P01	V-slot 20x20 aluminium extrusion	21
P04	Bed rollers	6
P05	Roller brackets (3D printed)	12
P06	Bed riser blocks (3D printed)	6
P07	Outfeed table board	1
P08	Outfeed table risers (3D printed)	3
P09	Scrap tray (3D printed)	1
P10	Mini V-wheel gantry plate kits	46
P11	NEMA 17 stepper motor	4
P12	NEMA 17 with integrated T8 lead screw	4
P13	Z motor bracket (3D printed)	2
P14	Lead-screw nut block (3D printed)	2
P17	5 mm cross shafts + bearings	2
P22	MeArm-type 4-servo arm kit	8
P23	Arm mount plate (3D printed)	2
P26	Cutter 'torch' for the prototype (pen holder)	1
P27	Handler magnet	1
P28	Model beams (3D printed)	1
P29	USB wide-angle camera (safety zone)	1
P30	CSI camera (close-up of the Cutter)	1
P32	Emergency stop station	2
P34	Door interlock switch	1
P38	IR break-beam sensor pair	2
P39	Polycarbonate sheet 2 mm	1
P44	Control box	1
P48	VL53L1X time-of-flight distance sensor	1
P49	Pen touch-off micro-switch	1
P50	Load cell 1 kg + HX711	1
P52	IR flame sensor module	1
P53	Line laser module (Class 2)	1

Steel Beam Cutting Cell - 1:10 prototype
Sheet: general assembly

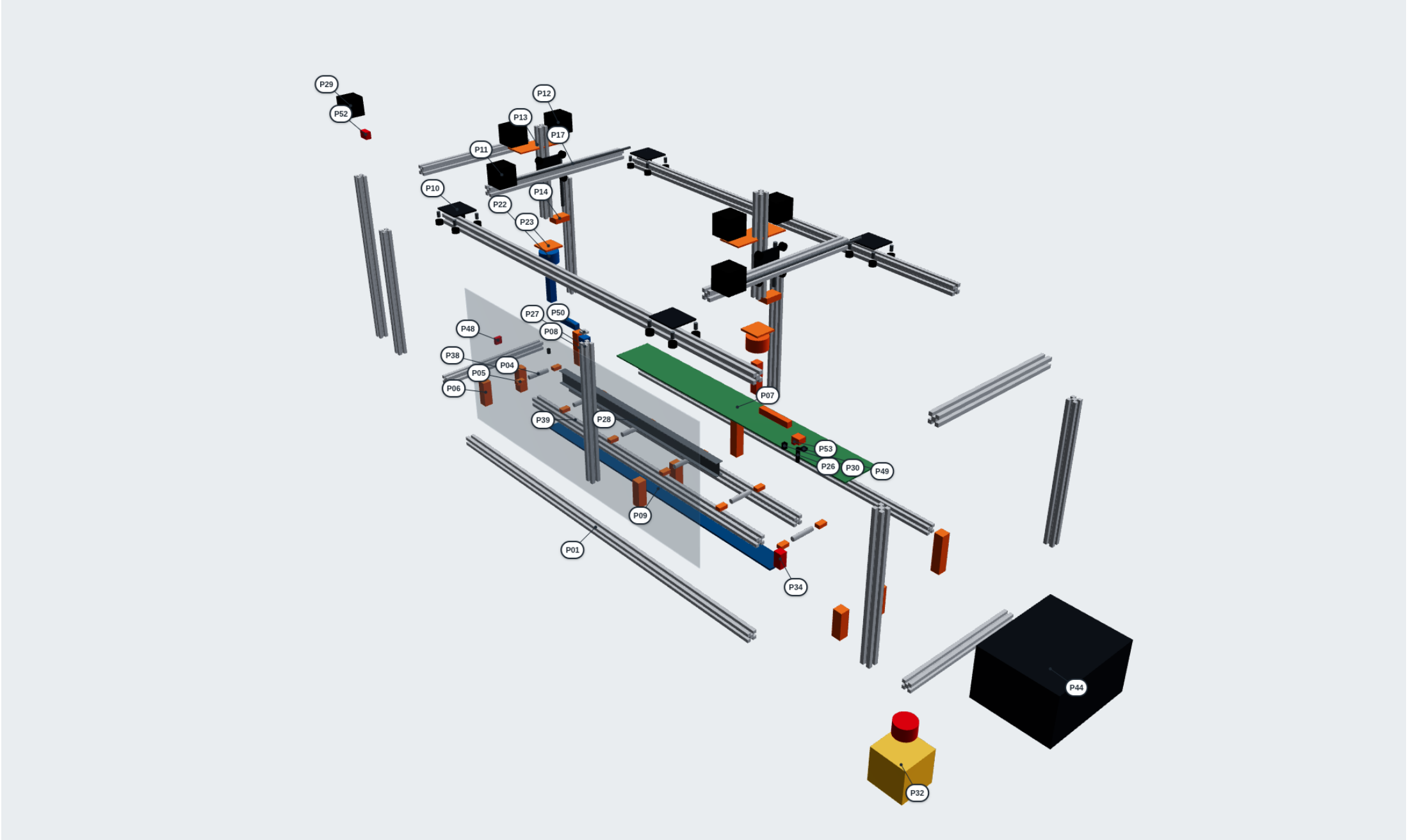
Drawing: BC-P1-GA

Drawn: Ravi Mahadeva
Date: 05 Oct 2026 Units: mm Scale: not to scale

Sheet 1
CAD: cad/prototype_1to10.step

Exploded view - what goes where

The parts pulled apart from the middle so you can see how they fit together.



Part	Name	No.
P01	V-slot 20x20 aluminium extrusion	21
P04	Bed rollers	6
P05	Roller brackets (3D printed)	12
P06	Bed riser blocks (3D printed)	6
P07	Outfeed table board	1
P08	Outfeed table risers (3D printed)	3
P09	Scrap tray (3D printed)	1
P10	Mini V-wheel gantry plate kits	46
P11	NEMA 17 stepper motor	4
P12	NEMA 17 with integrated T8 lead screw	4
P13	Z motor bracket (3D printed)	2
P14	Lead-screw nut block (3D printed)	2
P17	5 mm cross shafts + bearings	2
P22	MeArm-type 4-servo arm kit	8
P23	Arm mount plate (3D printed)	2
P26	Cutter 'torch' for the prototype (pen holder)	1
P27	Handler magnet	1
P28	Model beams (3D printed)	1
P29	USB wide-angle camera (safety zone)	1
P30	CSI camera (close-up of the Cutter)	1
P32	Emergency stop station	2
P34	Door interlock switch	1
P38	IR break-beam sensor pair	2
P39	Polycarbonate sheet 2 mm	1
P44	Control box	1
P48	VL53L1X time-of-flight distance sensor	1
P49	Pen touch-off micro-switch	1
P50	Load cell 1 kg + HX711	1
P52	IR flame sensor module	1
P53	Line laser module (Class 2)	1

Steel Beam Cutting Cell - 1:10 prototype

Sheet: exploded view

Drawing: BC-P1-GA

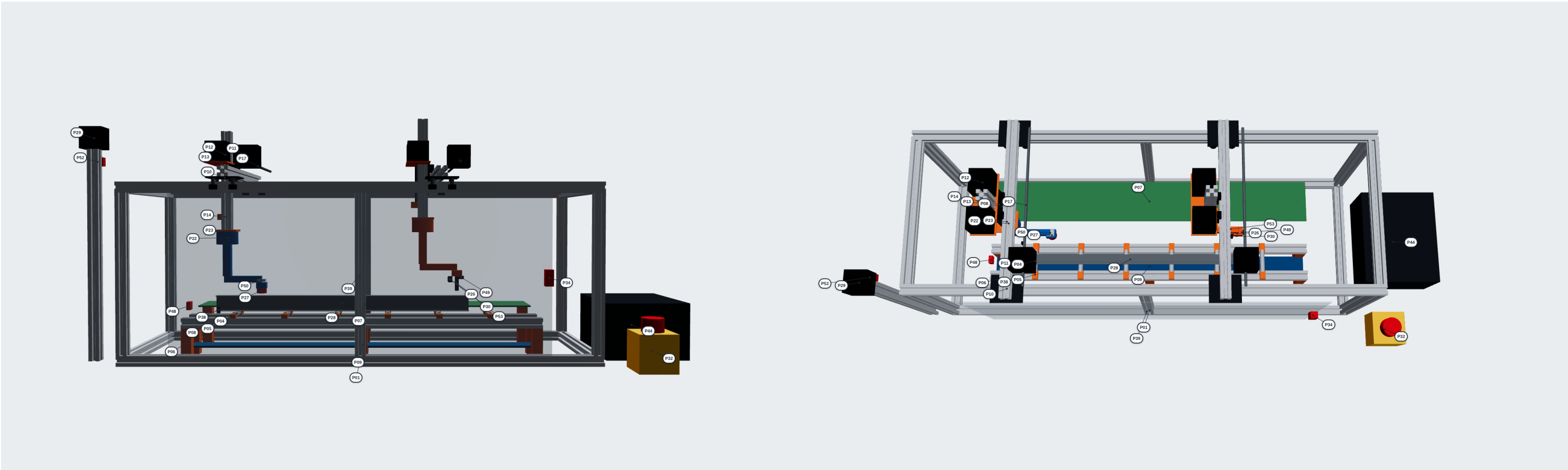
Drawn: Ravi Mahadeva

Date: 05 Oct 2026 Units: mm Scale: not to scale

Sheet 2

CAD: cad/prototype_1to10.step

Front view and top view



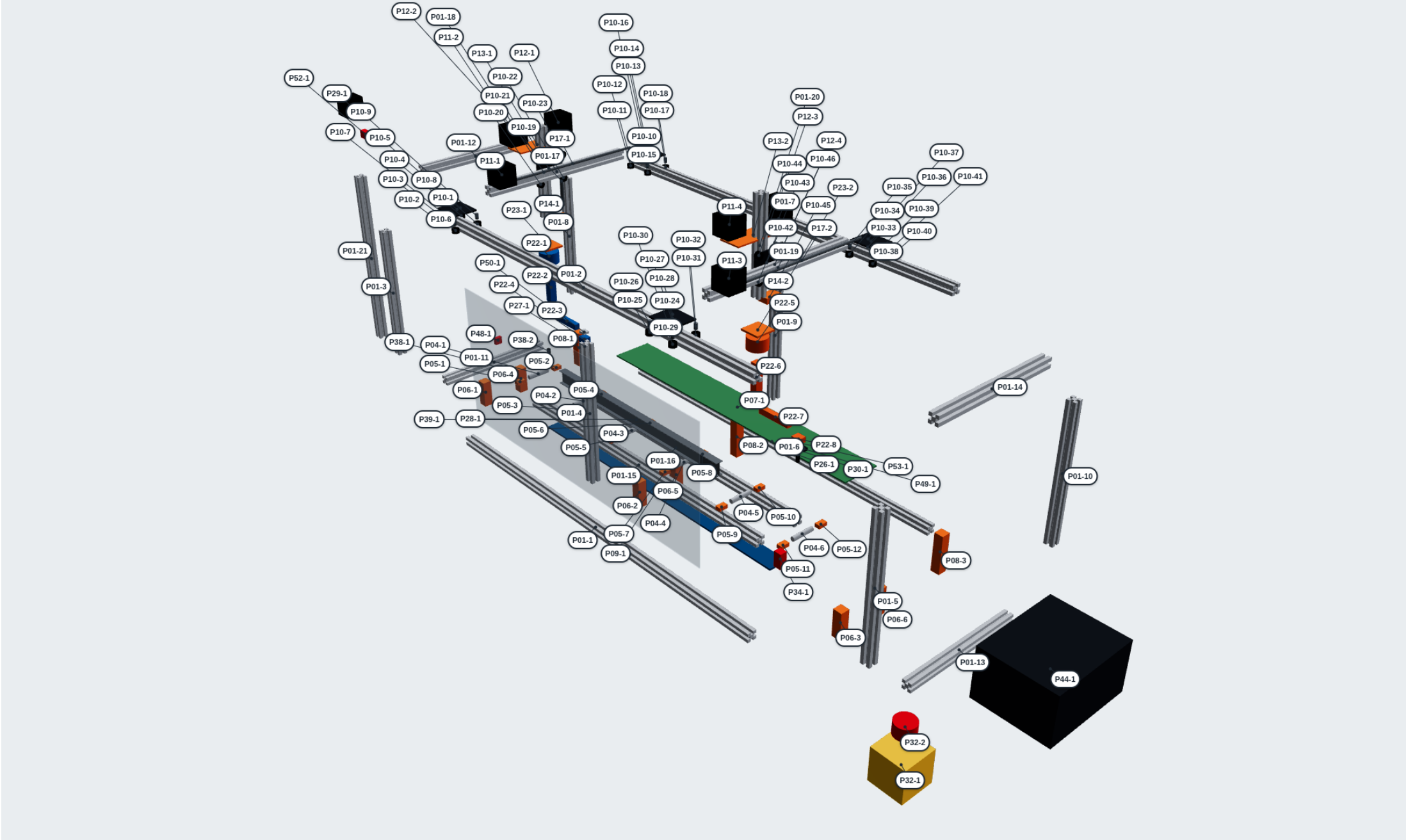
Front - from the operator's side

Top - looking down

Steel Beam Cutting Cell - 1:10 prototype Sheet: front and top views	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 05 Oct 2026 Units: mm Scale: not to scale	Sheet 3 CAD: cad/prototype_1to10.step

Every position

Each balloon is a position: part number - copy (P11-2 = the second NEMA 17 motor). The list on the following sheets says what each position is and where it goes.



Steel Beam Cutting Cell - 1:10 prototype Sheet: every position	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 05 Oct 2026 Units: mm Scale: not to scale	Sheet 4 CAD: cad/prototype_1to10.step

Parts list - part numbers, positions and where to buy

Part	Name	What to look for	Positions in the model	Buy	Each	Where to buy
P01	V-slot 20x20 aluminium extrusion	cut to size - lengths in cad/prototype_cut_list.csv (frame, bed rails, bridges, Z axes, camera post)	P01-1, P01-2, P01-3, P01-4, P01-5, P01-6, P01-7, P01-8, P01-9, P01-10, P01-11, P01-12, P01-13, P01-14, P01-15, P01-16, P01-17, P01-18, P01-19, P01-20, P01-21	9.5 m	£8	Ooznest (cut to size), Amazon UK search shop
P02	Corner brackets + M5 T-nuts + M5 bolts	30 cast 20-series corner brackets, 120 drop-in T-nuts, M5x8 button heads	not drawn (wiring, inside the control box, tools)	1 kit	£30	Ooznest, Amazon UK search shop
P03	Rubber feet	stick-on rubber feet 20 mm	not drawn (wiring, inside the control box, tools)	4	£1	Amazon UK search shop
P04	Bed rollers	10 mm printed sleeves on 3 mm silver steel rod (50 mm), 2 x 623ZZ bearings each	P04-1, P04-2, P04-3, P04-4, P04-5, P04-6	6	£2	Amazon UK, Simply Bearings search shop
P05	Roller brackets (3D printed)	PETG, clip on the 20x20 bed rails	P05-1, P05-2, P05-3, P05-4, P05-5, P05-6, P05-7, P05-8, P05-9, P05-10, P05-11, P05-12	12	£0	print make / print it
P06	Bed riser blocks (3D printed)	PETG blocks that hold the bed rails 60 mm above the base	P06-1, P06-2, P06-3, P06-4, P06-5, P06-6	6	£0	print make / print it
P07	Outfeed table board	3 mm plywood strip 700 x 90 mm	P07-1	1	£4	builders' merchant, Amazon UK search shop
P08	Outfeed table risers (3D printed)	PETG	P08-1, P08-2, P08-3	3	£0	print make / print it
P09	Scrap tray (3D printed)	700 x 44 mm tray under the bed	P09-1	1	£0	print make / print it
P10	Mini V-wheel gantry plate kits	20-series mini gantry plate + 4 mini V-wheels (625 bearings) + eccentric spacers	P10-1, P10-2, P10-3, P10-4, P10-5, P10-6, P10-7, P10-8, P10-9, P10-10, P10-11, P10-12, P10-13, P10-14, P10-15, P10-16, P10-17, P10-18, P10-19, P10-20, P10-21, P10-22, P10-23, P10-24, P10-25, P10-26, P10-27, P10-28, P10-29, P10-30, P10-31, P10-32, P10-33, P10-34, P10-35, P10-36, P10-37, P10-38, P10-39, P10-40, P10-41, P10-42, P10-43, P10-44, P10-45, P10-46	6	£12	Ooznest search shop
P11	NEMA 17 stepper motor	1.5-2 A, 40 mm body, 5 mm shaft (a NEMA 14 is lighter if you prefer)	P11-1, P11-2, P11-3, P11-4	4	£12	Ooznest, StepperOnline search shop
P12	NEMA 17 with integrated T8 lead screw	T8 x 2 mm lead, 150 mm screw, with brass nut	P12-1, P12-2, P12-3, P12-4	2	£22	StepperOnline, Amazon UK search shop
P13	Z motor bracket (3D printed)	PETG shelf on top of the carriage plate, holds the Z and Y motors	P13-1, P13-2	2	£0	print make / print it
P14	Lead-screw nut block (3D printed)	PETG block with the brass T8 nut, fixed to the Z axis	P14-1, P14-2	2	£0	print make / print it
P15	GT2 belt 6 mm	5 m (2 per bridge along the rails, 1 per carriage)	not drawn (wiring, inside the control box, tools)	1	£8	Ooznest search shop
P16	GT2 16T pulleys + idlers	4 x 16T 5 mm bore (cross shafts), 2 x 16T (Y), 8 idlers	not drawn (wiring, inside the control box, tools)	1 set	£20	Ooznest, Amazon UK search shop
P17	5 mm cross shafts + bearings	2 x 360 mm 5 mm silver steel, 4 x 625ZZ in printed blocks, 2 x 5-5 mm couplers	P17-1, P17-2	1 set	£12	Amazon UK, Simply Bearings search shop
P18	Home / limit switches	mechanical microswitch with lever, NC wiring	not drawn (wiring, inside the control box, tools)	8	£1.50	Amazon UK search shop
P19	FluidNC 6-axis CNC controller (ESP32)	6 driver sockets, USB to the Jetson (e.g. Elecrow / Bart Dring 6x CNC Controller)	not drawn (wiring, inside the control box, tools)	1	£40	Elecrow, Tindie product page
P20	TMC2209 stepper driver modules	StepStick format	not drawn (wiring, inside the control box, tools)	6	£5	Amazon UK, Ooznest search shop
P21	Motor cable + drag chain	5 m 4-core 0.25 mm2, 1 m 7x7 drag chain	not drawn (wiring, inside the control box, tools)	1 set	£15	Amazon UK search shop
P22	MeArm-type 4-servo arm kit	small 4-servo arm with MG90S metal-gear micro servos (reach ~150 mm); swap the gripper for the pen / magnet	P22-1, P22-2, P22-3, P22-4, P22-5, P22-6, P22-7, P22-8	2	£37	Cool Components, Kitronik product page
P23	Arm mount plate (3D printed)	PETG plate: bolts the arm base under the Z axis	P23-1, P23-2	2	£0	print make / print it
P24	PCA9685 16-channel servo board	I2C - wired straight to the Jetson's 40-pin header (pins 3 / 5), drives all 8 servos	not drawn (wiring, inside the control box, tools)	1	£6	Amazon UK, kunkune product page
P25	5 V power supply for the servos	Mean Well LRS-35-5 (5 V 7 A) - servos and magnet	not drawn (wiring, inside the control box, tools)	1	£12	RS, Farnell search shop

Steel Beam Cutting Cell - 1:10 prototype Sheet: parts list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 05 Oct 2026 Units: mm Scale: not to scale	Sheet 5 CAD: cad/prototype_1to10.step

Part	Name	What to look for	Positions in the model	Buy	Each	Where to buy
P26	Cutter 'torch' for the prototype (pen holder)	fine-liner pen in a sprung 3D-printed holder (marks every cut line on the model beam)	P26-1	1	£2	print <i>make / print it</i>
P27	Handler magnet	5 V 20 mm lifting electromagnet (~2.5 kg hold) + MOSFET module + flyback diode	P27-1	1	£8	Amazon UK search shop
P28	Model beams (3D printed)	UB 305x165 at 1:10 (30 x 16 mm), 500 mm long, with steel tape on the top flange	P28-1	4	£0	print <i>make / print it</i>
P29	USB wide-angle camera (safety zone)	1080p UVC, 90-120 deg view (e.g. Logitech C920/C922 or an Arducam USB wide-angle)	P29-1	1	£50	Amazon UK, Pimoroni search shop
P30	CSI camera (close-up of the Cutter)	Arducam IMX219 for Jetson Orin Nano (22-pin cable), wide-angle lens	P30-1	1	£25	Arducam, RobotShop UK product page
P31	Powered USB 3 hub	4-port, own power supply	<i>not drawn (wiring, inside the control box, tools)</i>	1	£20	Amazon UK search shop
P32	Emergency stop station	yellow box, red 40 mm mushroom, turn to release, 2 NC contacts (e.g. Schneider Harmony XALK178 family)	P32-1, P32-2	1	£42	RS, Farnell, TME search shop
P33	Safety relay	Pilz PNOZ s3 24 V DC (dual channel, manual reset) or equivalent	<i>not drawn (wiring, inside the control box, tools)</i>	1	£180	RS, Farnell (new) / ~£75 used search shop
P34	Door interlock switch	non-contact coded magnetic safety switch, NC (e.g. Pilz PSEN or Schmersal BNS)	P34-1	1	£60	RS, Farnell search shop
P35	Reset push button	22 mm blue illuminated, 1 NO	<i>not drawn (wiring, inside the control box, tools)</i>	1	£8	RS, Amazon UK search shop
P36	Stack light	24 V mini LED tower, red / amber / green	<i>not drawn (wiring, inside the control box, tools)</i>	1	£15	Amazon UK search shop
P37	4-channel relay module (opto-isolated)	3.3 V logic input, 24 V contacts	<i>not drawn (wiring, inside the control box, tools)</i>	1	£8	Amazon UK search shop
P38	IR break-beam sensor pair	5 mm IR break-beam (demonstrates the light-curtain input)	P38-1, P38-2	2	£6	Pimoroni, The Pi Hut search shop
P39	Polycarbonate sheet 2 mm	front door, ends and lid (~0.6 m2) - polycarbonate, not acrylic (acrylic shatters)	P39-1	0.6 m2	£35	plastics stockist, Amazon UK search shop
P40	Hinges + handle	2 small hinges, 1 handle (20-series)	<i>not drawn (wiring, inside the control box, tools)</i>	1 set	£12	Ooznest, Amazon UK search shop
P41	24 V power supply	Mean Well LRS-100-24 (24 V 4.5 A) - motors, controller, safety circuit	<i>not drawn (wiring, inside the control box, tools)</i>	1	£14	RS, Farnell search shop
P42	IEC mains inlet with switch and fuse	panel mount, 3 A fuse	<i>not drawn (wiring, inside the control box, tools)</i>	1	£8	RS, Farnell search shop
P43	DIN rail + terminals + fuse holders	short 35 mm DIN rail, 12 terminals, 3 x 5x20 fuse holders	<i>not drawn (wiring, inside the control box, tools)</i>	1 set	£20	RS, Farnell search shop
P44	Control box	ABS enclosure ~200 x 160 x 110 with cable glands	P44-1	1	£20	RS, CPC search shop
P45	Wire + ferrules + crimp tool	0.5 and 0.75 mm2 flexible, ferrule kit	<i>not drawn (wiring, inside the control box, tools)</i>	1 set	£25	Amazon UK, CPC search shop
P46	Jetson Orin Nano Super (4 GB)	you already have it - docs/JETSON_SPECS.md	<i>not drawn (wiring, inside the control box, tools)</i>	1	£0	- <i>make / print it</i>
P47	PETG and PLA filament	1 kg each	<i>not drawn (wiring, inside the control box, tools)</i>	2	£20	Amazon UK, 3DJake search shop
P48	VL53L1X time-of-flight distance sensor	I2C breakout, up to 4 m - the datum sensor: finds where the model beam starts	P48-1	1	£12	Pimoroni, The Pi Hut search shop
P49	Pen touch-off micro-switch	micro-switch with roller lever, on the sprung pen holder - finds the beam surface (the prototype's initial height sensing)	P49-1	2	£1.50	Amazon UK, CPC search shop
P50	Load cell 1 kg + HX711	bar load cell 1 kg with HX711 amplifier - between the Handler's wrist and magnet: weighs the part (slipping, wrong part, not cut free)	P50-1	1	£7	Amazon UK, The Pi Hut search shop
P51	INA219 current sensor	I2C current sensor in the magnet supply - checks the magnet is really on	<i>not drawn (wiring, inside the control box, tools)</i>	1	£5	Pimoroni, Amazon UK search shop
P52	IR flame sensor module	IR flame sensor with digital output - demonstrates the fire detector	P52-1	1	£3	Amazon UK search shop

Steel Beam Cutting Cell - 1:10 prototype Sheet: position list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 05 Oct 2026 Units: mm Scale: not to scale	Sheet 6 CAD: cad/prototype_1to10.step

Part	Name	What to look for	Positions in the model	Buy	Each	Where to buy
P53	Line laser module (Class 2)	650 nm line laser, Class 1 or 2 (1 mW or less) - with camera 2 it shows the beam's real profile	P53-1	1	£6	Amazon UK search shop

Steel Beam Cutting Cell - 1:10 prototype Sheet: position list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 05 Oct 2026 Units: mm Scale: not to scale	Sheet 7 CAD: cad/prototype_1to10.step

Position list - what each position is and where it goes

x is along the machine (infeed end = 0), y across (front, the operator's side, is negative), z up from the table. Sizes in mm, centre of each part.

Position	Part	What it is	Where	x	y	z
P01-1	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - base rail 900 mm	front (operator side), middle, 10 mm up	300	-150	10
P01-2	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - gantry rail 900 mm (the bridges' wheels run on it)	front (operator side), middle, 330 mm up	300	-150	330
P01-3	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - upright 300 mm	front (operator side), infeed end, 170 mm up	-140	-150	170
P01-4	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - upright 300 mm	front (operator side), middle, 170 mm up	300	-150	170
P01-5	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - upright 300 mm	front (operator side), middle, 170 mm up	740	-150	170
P01-6	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - base rail 900 mm	back, middle, 10 mm up	300	150	10
P01-7	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - gantry rail 900 mm (the bridges' wheels run on it)	back, middle, 330 mm up	300	150	330
P01-8	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - upright 300 mm	back, infeed end, 170 mm up	-140	150	170
P01-9	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - upright 300 mm	back, middle, 170 mm up	300	150	170
P01-10	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - upright 300 mm	back, middle, 170 mm up	740	150	170
P01-11	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - cross member 280 mm	middle, infeed end, 10 mm up	-140	0	10
P01-12	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - cross member 280 mm	middle, infeed end, 330 mm up	-140	0	330
P01-13	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - cross member 280 mm	middle, middle, 10 mm up	740	0	10
P01-14	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - cross member 280 mm	middle, middle, 330 mm up	740	0	330
P01-15	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - bed rail 700 mm	front (operator side), middle, 70 mm up	300	-80	70
P06-1	Bed riser blocks (3D printed)	Bed riser block (3D printed) - holds the bed rails on the base	front (operator side), infeed end, 30 mm up	-40	-80	30
P06-2	Bed riser blocks (3D printed)	Bed riser block (3D printed) - holds the bed rails on the base	front (operator side), middle, 30 mm up	300	-80	30
P06-3	Bed riser blocks (3D printed)	Bed riser block (3D printed) - holds the bed rails on the base	front (operator side), middle, 30 mm up	640	-80	30
P01-16	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - bed rail 700 mm	middle, middle, 70 mm up	300	-20	70
P06-4	Bed riser blocks (3D printed)	Bed riser block (3D printed) - holds the bed rails on the base	middle, infeed end, 30 mm up	-40	-20	30
P06-5	Bed riser blocks (3D printed)	Bed riser block (3D printed) - holds the bed rails on the base	middle, middle, 30 mm up	300	-20	30
P06-6	Bed riser blocks (3D printed)	Bed riser block (3D printed) - holds the bed rails on the base	middle, middle, 30 mm up	640	-20	30
P04-1	Bed rollers	Bed roller: 10 mm printed sleeve on a 3 mm steel rod, 2 x 623 bearings	middle, infeed end, 85 mm up	50	-50	85
P05-1	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	front (operator side), infeed end, 84 mm up	50	-80	84
P05-2	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	middle, infeed end, 84 mm up	50	-20	84
P04-2	Bed rollers	Bed roller: 10 mm printed sleeve on a 3 mm steel rod, 2 x 623 bearings	middle, infeed end, 85 mm up	150	-50	85
P05-3	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	front (operator side), infeed end, 84 mm up	150	-80	84
P05-4	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	middle, infeed end, 84 mm up	150	-20	84
P04-3	Bed rollers	Bed roller: 10 mm printed sleeve on a 3 mm steel rod, 2 x 623 bearings	middle, middle, 85 mm up	250	-50	85
P05-5	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	front (operator side), middle, 84 mm up	250	-80	84
P05-6	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	middle, middle, 84 mm up	250	-20	84
P04-4	Bed rollers	Bed roller: 10 mm printed sleeve on a 3 mm steel rod, 2 x 623 bearings	middle, middle, 85 mm up	350	-50	85
P05-7	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	front (operator side), middle, 84 mm up	350	-80	84
P05-8	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	middle, middle, 84 mm up	350	-20	84
P04-5	Bed rollers	Bed roller: 10 mm printed sleeve on a 3 mm steel rod, 2 x 623 bearings	middle, middle, 85 mm up	450	-50	85
P05-9	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	front (operator side), middle, 84 mm up	450	-80	84
P05-10	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	middle, middle, 84 mm up	450	-20	84
P04-6	Bed rollers	Bed roller: 10 mm printed sleeve on a 3 mm steel rod, 2 x 623 bearings	middle, middle, 85 mm up	550	-50	85
P05-11	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	front (operator side), middle, 84 mm up	550	-80	84
P05-12	Roller brackets (3D printed)	Roller bracket (3D printed, PETG) - clips on the bed rail	middle, middle, 84 mm up	550	-20	84
P09-1	Scrap tray (3D printed)	Scrap tray (3D printed) - offcuts fall into it	middle, middle, 22 mm up	300	-50	22
P07-1	Outfeed table board	Outfeed table: 3 mm plywood strip on printed risers	back, middle, 88 mm up	300	85	88
P08-1	Outfeed table risers (3D printed)	Outfeed table riser (3D printed)	back, infeed end, 44 mm up	-40	85	44
P08-2	Outfeed table risers (3D printed)	Outfeed table riser (3D printed)	back, middle, 44 mm up	300	85	44
P08-3	Outfeed table risers (3D printed)	Outfeed table riser (3D printed)	back, middle, 44 mm up	640	85	44
P28-1	Model beams (3D printed)	Model beam: UB 305x165x40 at 1:10 (30 x 16 mm), 3D printed, steel tape on top	middle, middle, 105 mm up	260	-50	105

Steel Beam Cutting Cell - 1:10 prototype Sheet: position list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 05 Oct 2026 Units: mm Scale: not to scale	Sheet 8 CAD: cad/prototype_1to10.step

Position	Part	What it is	Where	x	y	z
P01-17	V-slot 20x20 aluminium extrusion	Handler bridge: 20x20 V-slot aluminium extrusion 340 mm	middle, infeed end, 359 mm up	50	0	359
P10-1	Mini V-wheel gantry plate kits	Handler bridge end: mini V-wheel gantry plate (20 series), clamps the gantry rail	front (operator side), infeed end, 348 mm up	50	-150	348
P10-2	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	front (operator side), infeed end, 330 mm up	32	-166	330
P10-3	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	front (operator side), infeed end, 340 mm up	32	-166	340
P10-4	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	front (operator side), infeed end, 330 mm up	32	-134	330
P10-5	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	front (operator side), infeed end, 340 mm up	32	-134	340
P10-6	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	front (operator side), infeed end, 330 mm up	68	-166	330
P10-7	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	front (operator side), infeed end, 340 mm up	68	-166	340
P10-8	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	front (operator side), infeed end, 330 mm up	68	-134	330
P10-9	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	front (operator side), infeed end, 340 mm up	68	-134	340
P10-10	Mini V-wheel gantry plate kits	Handler bridge end: mini V-wheel gantry plate (20 series), clamps the gantry rail	back, infeed end, 348 mm up	50	150	348
P10-11	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	back, infeed end, 330 mm up	32	134	330
P10-12	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	back, infeed end, 340 mm up	32	134	340
P10-13	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	back, infeed end, 330 mm up	32	166	330
P10-14	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	back, infeed end, 340 mm up	32	166	340
P10-15	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	back, infeed end, 330 mm up	68	134	330
P10-16	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	back, infeed end, 340 mm up	68	134	340
P10-17	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	back, infeed end, 330 mm up	68	166	330
P10-18	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	back, infeed end, 340 mm up	68	166	340
P11-1	NEMA 17 stepper motor	Handler X motor: NEMA 17 driving both rails through a 5 mm cross shaft and GT2 belts	front (operator side), infeed end, 390 mm up	85	-105	390
P17-1	5 mm cross shafts + bearings	5 mm cross shaft with GT2 16T pulleys at both ends	middle, infeed end, 379 mm up	85	0	379
P10-19	Mini V-wheel gantry plate kits	Handler carriage: mini V-wheel gantry plate on the side of the bridge (Y axis)	middle, infeed end, 359 mm up	32	0	359
P10-20	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) - runs in the bridge's top / bottom slot	middle, infeed end, 343 mm up	38	-18	343
P10-21	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) - runs in the bridge's top / bottom slot	middle, infeed end, 375 mm up	38	-18	375
P10-22	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) - runs in the bridge's top / bottom slot	middle, infeed end, 343 mm up	38	18	343
P10-23	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) - runs in the bridge's top / bottom slot	middle, infeed end, 375 mm up	38	18	375
P13-1	Z motor bracket (3D printed)	Handler motor shelf (3D printed): holds the Z and Y motors on top of the carriage plate	middle, infeed end, 384 mm up	10	0	384
P11-2	NEMA 17 stepper motor	Handler Y motor: NEMA 17 + GT2 belt along the bridge	middle, infeed end, 406 mm up	10	-35	406
P12-1	NEMA 17 with integrated T8 lead screw	Handler Z motor: NEMA 17 with integrated T8 x 2 mm lead screw (150 mm)	middle, infeed end, 406 mm up	10	35	406
P01-18	V-slot 20x20 aluminium extrusion	Handler Z axis: 20x20 V-slot aluminium extrusion 200 mm on a T8 lead screw (slides on the carriage plate)	middle, infeed end, 350 mm up	21	0	350
P12-2	NEMA 17 with integrated T8 lead screw	T8 x 2 mm lead screw, 150 mm	middle, infeed end, 327 mm up	10	35	327
P14-1	Lead-screw nut block (3D printed)	Handler lead-screw nut block on the Z axis (3D printed + brass T8 nut)	middle, infeed end, 276 mm up	10	28	276
P23-1	Arm mount plate (3D printed)	Handler arm mount plate (3D printed)	middle, infeed end, 248 mm up	21	0	248
P22-1	MeArm-type 4-servo arm kit	Handler arm: MeArm-type 4-servo arm kit (MG90S servos) - base	middle, infeed end, 233 mm up	21	0	233
P22-2	MeArm-type 4-servo arm kit	Handler arm: upper arm (~80 mm)	middle, infeed end, 188 mm up	21	0	188
P22-3	MeArm-type 4-servo arm kit	Handler arm: forearm (~80 mm)	middle, infeed end, 149 mm up	53	0	149
P22-4	MeArm-type 4-servo arm kit	Handler arm: wrist / tool holder	middle, infeed end, 136 mm up	91	0	136
P27-1	Handler magnet	Handler magnet: 5 V 20 mm lifting electromagnet (holds about 2.5 kg)	middle, infeed end, 125 mm up	91	0	125
P01-19	V-slot 20x20 aluminium extrusion	Cutter bridge: 20x20 V-slot aluminium extrusion 340 mm	middle, middle, 359 mm up	450	0	359
P10-24	Mini V-wheel gantry plate kits	Cutter bridge end: mini V-wheel gantry plate (20 series), clamps the gantry rail	front (operator side), middle, 348 mm up	450	-150	348
P10-25	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	front (operator side), middle, 330 mm up	432	-166	330
P10-26	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	front (operator side), middle, 340 mm up	432	-166	340
P10-27	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	front (operator side), middle, 330 mm up	432	-134	330
P10-28	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	front (operator side), middle, 340 mm up	432	-134	340
P10-29	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	front (operator side), middle, 330 mm up	468	-166	330
P10-30	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	front (operator side), middle, 340 mm up	468	-166	340
P10-31	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	front (operator side), middle, 330 mm up	468	-134	330
P10-32	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	front (operator side), middle, 340 mm up	468	-134	340
P10-33	Mini V-wheel gantry plate kits	Cutter bridge end: mini V-wheel gantry plate (20 series), clamps the gantry rail	back, middle, 348 mm up	450	150	348

Steel Beam Cutting Cell - 1:10 prototype Sheet: position list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 05 Oct 2026 Units: mm Scale: not to scale	Sheet 9 CAD: cad/prototype_1to10.step

Position	Part	What it is	Where	x	y	z
P10-34	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	back, middle, 330 mm up	432	134	330
P10-35	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	back, middle, 340 mm up	432	134	340
P10-36	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	back, middle, 330 mm up	432	166	330
P10-37	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	back, middle, 340 mm up	432	166	340
P10-38	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	back, middle, 330 mm up	468	134	330
P10-39	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	back, middle, 340 mm up	468	134	340
P10-40	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) on an eccentric spacer - runs in the rail's side slot	back, middle, 330 mm up	468	166	330
P10-41	Mini V-wheel gantry plate kits	Eccentric spacer + M5 bolt: holds the V-wheel under the gantry plate	back, middle, 340 mm up	468	166	340
P11-3	NEMA 17 stepper motor	Cutter X motor: NEMA 17 driving both rails through a 5 mm cross shaft and GT2 belts	front (operator side), middle, 390 mm up	485	-105	390
P17-2	5 mm cross shafts + bearings	5 mm cross shaft with GT2 16T pulleys at both ends	middle, middle, 379 mm up	485	0	379
P10-42	Mini V-wheel gantry plate kits	Cutter carriage: mini V-wheel gantry plate on the side of the bridge (Y axis)	middle, middle, 359 mm up	432	0	359
P10-43	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) - runs in the bridge's top / bottom slot	middle, middle, 343 mm up	438	-18	343
P10-44	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) - runs in the bridge's top / bottom slot	middle, middle, 375 mm up	438	-18	375
P10-45	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) - runs in the bridge's top / bottom slot	middle, middle, 343 mm up	438	18	343
P10-46	Mini V-wheel gantry plate kits	Mini V-wheel (625 bearing) - runs in the bridge's top / bottom slot	middle, middle, 375 mm up	438	18	375
P13-2	Z motor bracket (3D printed)	Cutter motor shelf (3D printed): holds the Z and Y motors on top of the carriage plate	middle, middle, 384 mm up	410	0	384
P11-4	NEMA 17 stepper motor	Cutter Y motor: NEMA 17 + GT2 belt along the bridge	middle, middle, 406 mm up	410	-35	406
P12-3	NEMA 17 with integrated T8 lead screw	Cutter Z motor: NEMA 17 with integrated T8 x 2 mm lead screw (150 mm)	middle, middle, 406 mm up	410	35	406
P01-20	V-slot 20x20 aluminium extrusion	Cutter Z axis: 20x20 V-slot aluminium extrusion 200 mm on a T8 lead screw (slides on the carriage plate)	middle, middle, 375 mm up	421	0	375
P12-4	NEMA 17 with integrated T8 lead screw	T8 x 2 mm lead screw, 150 mm	middle, middle, 339 mm up	410	35	339
P14-2	Lead-screw nut block (3D printed)	Cutter lead-screw nut block on the Z axis (3D printed + brass T8 nut)	middle, middle, 301 mm up	410	28	301
P23-2	Arm mount plate (3D printed)	Cutter arm mount plate (3D printed)	middle, middle, 273 mm up	421	0	273
P22-5	MeArm-type 4-servo arm kit	Cutter arm: MeArm-type 4-servo arm kit (MG90S servos) - base	middle, middle, 258 mm up	421	0	258
P22-6	MeArm-type 4-servo arm kit	Cutter arm: upper arm (~80 mm)	middle, middle, 213 mm up	421	0	213
P22-7	MeArm-type 4-servo arm kit	Cutter arm: forearm (~80 mm)	middle, middle, 174 mm up	453	0	174
P22-8	MeArm-type 4-servo arm kit	Cutter arm: wrist / tool holder	middle, middle, 161 mm up	491	0	161
P26-1	Cutter 'torch' for the prototype (pen holder)	Cutter 'torch' for the prototype: fine-liner pen in a sprung 3D-printed holder (marks the cut lines)	middle, middle, 140 mm up	491	0	140
P44-1	Control box	Control box: Jetson Orin Nano, FluidNC 6-axis board, PCA9685 servo board, 24 V and 5 V supplies, fuses, safety relay	middle, middle, 55 mm up	850	0	55
P32-1	Emergency stop station	Emergency stop: 40 mm red mushroom, 2 NC contacts, yellow box (BS EN ISO 13850)	front (operator side), middle, 35 mm up	825	-195	35
P32-2	Emergency stop station	Emergency stop head	front (operator side), middle, 82 mm up	825	-195	82
P39-1	Polycarbonate sheet 2 mm	Enclosure door: 2 mm polycarbonate on a 20x20 frame, with a safety interlock switch	front (operator side), middle, 170 mm up	300	-171	170
P34-1	Door interlock switch	Door interlock switch (magnetic coded, NC) - opens = protective stop	front (operator side), middle, 165 mm up	642	-179	165
P29-1	USB wide-angle camera (safety zone)	Camera: USB wide-angle (or IMX219 CSI) on a 20x20 post - YOLO watches the cell	front (operator side), infeed end, 420 mm up	-190	-150	420
P01-21	V-slot 20x20 aluminium extrusion	20x20 V-slot aluminium extrusion - camera post	front (operator side), infeed end, 205 mm up	-190	-150	205
P48-1	VL53L1X time-of-flight distance sensor	Datum sensor: VL53L1X time-of-flight at the bed's infeed end - finds where the model beam starts	middle, infeed end, 100 mm up	-44	-50	100
P38-1	IR break-beam sensor pair	Bar-present IR break-beam (one each side of the bed)	front (operator side), infeed end, 96 mm up	23	-88	96
P38-2	IR break-beam sensor pair	Bar-present IR break-beam (one each side of the bed)	middle, infeed end, 96 mm up	23	-12	96
P52-1	IR flame sensor module	IR flame sensor module (demo of the fire detector)	front (operator side), infeed end, 378 mm up	-190	-134	378
P50-1	Load cell 1 kg + HX711	Load cell (1 kg) between wrist and magnet - weighs the part: slipping, wrong part, not cut free	middle, infeed end, 147 mm up	91	0	147
P30-1	CSI camera (close-up of the Cutter)	Camera 2: IMX219 close-up beside the pen - the cut line, the bar edge	middle, middle, 149 mm up	477	-8	149
P53-1	Line laser module (Class 2)	Line laser (Class 2) - with camera 2 it shows the beam's real profile	middle, middle, 148 mm up	477	8	148
P49-1	Pen touch-off micro-switch	Pen touch-off micro-switch on the sprung holder - finds the beam's surface	middle, middle, 152 mm up	500	0	152

Steel Beam Cutting Cell - 1:10 prototype Sheet: position list	Drawing: BC-P1-GA
Drawn: Ravi Mahadeva Date: 05 Oct 2026 Units: mm Scale: not to scale	Sheet 10 CAD: cad/prototype_1to10.step